# LLM-Based Anomaly Detection in System Logs

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## Abstract

A milestone update will be given on how Large Language Models can be used to identify problems in system and network logs. The practical application that will be used is a web application designed using the MongoDB, Express, React, and Node (MERN) stack. The framework that will be used to scan the network logs is called LangChain combined with Claude. The topic number in this syllabus is: 14. LLM-Based Anomaly Detection in System Logs.

## CCS Concepts

• **Do Not Use This Code → Generate the Correct Terms for Your Paper**; *Generate the Correct Terms for Your Paper*; Generate the Correct Terms for Your Paper; Generate the Correct Terms for Your Paper.

## Keywords

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## 1 Introduction and Problem Statement

### 1.1 Motivation

The motivation for this project is to explore how LLMs can be used to identify problems in system and network logs, which can be a valuable tool for system administrators and security professionals. The significance of this project is that it can help to improve the efficiency and effectiveness of log analysis, which can lead to faster identification of issues and improved security. The objectives of this project are to develop a web application that can scan network logs using LangChain and Claude, and to evaluate the performance of the application in identifying anomalies in the logs.

### 1.2 Significance

### 1.3 Objectives

## 2 Related Work

This section contains relevant prior work with ACM-style citations.

## 3 Methodology

### 3.1 Techniques

### 3.2 Evaluation Metrics

## 4 Preliminary Results

Initial findings with brief analysis.

## 5 References

Proper ACM reference styles

## References

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